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IT 461: Practical Machine Learning
2ND Semester 1447 H

Alzheimer's Disease Stage Classification Using Brain MRI Images

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1. Introduction

Alzheimer's disease is a progressive neurological disorder and a leading cause of dementia globally. It is one of the leading causes of dementia worldwide, affecting millions of people. Early, accurate detection is essential for slowing progression and improving patient outcomes. While manual MRI diagnosis is often slow and subjective, machine learning (specifically deep learning) offers a faster, more consistent alternative for automated classification.

1.1 Background

Traditional approaches to Alzheimer's classification relied on manually extracted features combined with classical machine learning classifiers such as Support Vector Machines (SVM) and Random Forest. These approaches required expert-designed features and were limited in their ability to capture complex visual patterns. Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs), have improved performance by automatically learning hierarchical features directly from image pixels. Furthermore, transfer learning fine-tuning models pre-trained on large datasets like ImageNet enhances accuracy and reduces training time for smaller medical datasets.

1.2 Task Formulation

The task is a supervised binary image classification problem.

Input: A preprocessed grayscale brain MRI image resized to 224×224 pixels. The relevant features are the pixel intensity values that capture brain structure, including tissue density, ventricle size, and hippocampal volume.

Output: A binary label indicating the patient's condition:

- **0: Healthy** (Non-Demented)
- **1: Diseased** (Very Mild Demented, Mild Demented, or Moderate Demented)

To align the dataset with the classification task, the three dementia stages (Very Mild, Mild, and Moderate) were merged into a single 'Diseased' class.

Evaluation Metrics:

- **Accuracy:** Overall correctness of predictions
- **Precision:** Of all positive predictions, how many were correct
- **Recall:** Of all actual positives, how many were detected
- **F1-Score:** Harmonic mean of precision and recall
- **Confusion Matrix:** Visual breakdown of correct and incorrect predictions.

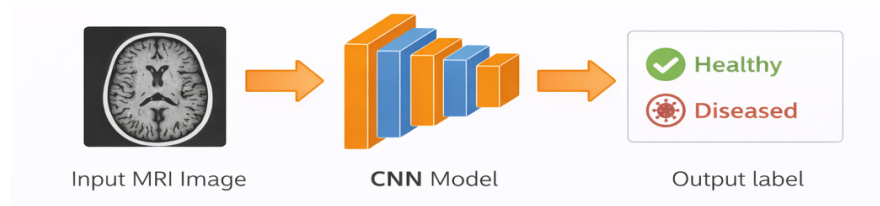


Figure 1: Task Illustration

2. Related Works

Several studies have demonstrated the effectiveness of deep learning for Alzheimer's disease classification using MRI images. These works provide benchmarks and methodological insights that inform our project.

2.1 Thevar et al. (2025): Comparative Analysis of ML Models

Thevar et al.[5]compared multiple machine learning models for classifying dementia stages using the OASIS dataset. They evaluated SVM, Random Forest, Logistic Regression, and CNN. Their results showed that CNN performed better than traditional machine learning models because it can automatically extract features from MRI images, leading to improved classification accuracy.

2.2 Hossain et al. (2024): Transfer Learning with Pre-trained CNNs

Hossain et al. [6] applied transfer learning for Alzheimer's stage classification using the ADNI MRI dataset. They tested VGG16, DenseNet121, and ResNet50 with 5-fold cross-validation. The results showed that VGG16 achieved the highest accuracy (95.0%), followed by DenseNet121 (90.4%), while ResNet50 performed lower (62.0%), highlighting the effectiveness of transfer learning.

2.3 Mohammed et al. (2021): Hybrid Deep Learning + SVM

Mohammed et al. [7] proposed a hybrid approach combining deep learning and traditional machine learning for Alzheimer's detection using the OASIS dataset. They used AlexNet and ResNet-50 for feature extraction and SVM for classification. The best results were achieved through AlexNet+SVM, reaching 94.8% accuracy and 99.70% AUC, showing the benefit of hybrid models.

2.4 Hu et al. (2025): Multi-task Learning with Feature Fusion

Hu et al.[8] developed a multi-task learning model for Alzheimer's classification using data from 2,956 patients across seven clinical centers. Their method combined radiomic features, deep learning features, and hippocampal segmentation, processed using LSTM with attention and classified with XGBoost. The model achieved 92.8% accuracy on external validation, demonstrating strong generalization.

Summary: These related works confirm the viability of CNN-based approaches for Alzheimer's classification, highlight the benefits of transfer learning and hybrid methods, and provide benchmark performance metrics against which our models can be compared.

3. Data

3.1 Dataset Description

The OASIS Alzheimer's Disease Classification dataset is a public medical image dataset sourced from Kaggle [1], curated for training and evaluating computer vision models focused on detecting and classifying stages of Alzheimer's disease. The dataset consists of preprocessed brain MRI images organized into labeled categories.

Source: Kaggle – OASIS Alzheimer Dataset

3.2 Dataset Characteristics

Characteristic	Value
Total Images	~10,400 MRI images
Total Unique Patients	68
Number of Classes	4
Classes	Non-Demented, Very Mild Demented, Mild Demented, Moderate Demented

Image Type	Brain MRI scans
Image Format	JPEG
Image Color	Grayscale
Task Type	Image Classification

Table 1: Dataset Characteristics

3.3 Class Distribution

Class	Approximate Count
Healthy	~3,000
Diseased	~7,400

Table 2: Class Distribution

The dataset is nearly balanced across the four original classes. However, after merging them into a binary classification problem (Healthy vs. Diseased), a mild class imbalance emerged.

3.4 Representative Examples

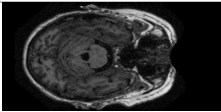
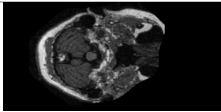
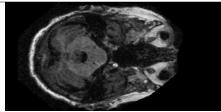
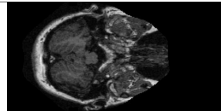
Class	Non Demented	Very Mild Demented	Mild Demented	Moderate Demented
Example				

Table 3: Representative Examples

3.5 Data Preprocessing

The following preprocessing steps were applied to all images:

Step	Description
Decoding	Raw JPEG images were read and decoded using TensorFlow's decode_jpeg
Resizing	All images were resized to 224×224 pixels to match pre-trained model input requirements
Normalization	Pixel values were scaled from [0, 255] to [0, 1] to stabilize and accelerate training

Table 4: Data Preprocessing

No additional cleaning was required as the dataset was already preprocessed.

3.6 Patient-Wise Grouping

To prevent data leakage, images were grouped by Patient ID. Patient IDs were extracted from filenames (e.g., "OAS1_0028_MR1_mpr-1_100.jpg" → patient ID "OAS1_0028"). All images from a single patient were kept within the same fold during cross-validation.

4. Methods

We used Convolutional Neural Networks (CNNs) with transfer learning as our primary methodology for this task.

Why

CNNs:

CNNs are the state-of-the-art for image analysis. Unlike traditional algorithms, they automatically learn hierarchical features from raw pixels. Lower layers detect simple patterns (edges, textures), while deeper layers identify complex, disease-relevant structures like tissue atrophy or ventricular enlargement. This end-to-end learning is essential for capturing the subtle, distributed biomarkers of Alzheimer's in MRI scans.

Why Transfer Learning:

Training a deep CNN from scratch requires millions of images to prevent overfitting. Our dataset (10,442 images from 68 patients) is sufficient for fine-tuning but small for training from scratch. Transfer learning overcomes this by leveraging features from a model pre-trained on **ImageNet** (14 million images). This approach provides three key benefits:

1. **Improved Performance:** The model begins with powerful, generalizable features instead of random weights.
2. **Reduced Training Time:** Convergence is faster since the model only requires fine-tuning.
3. **Built-in Regularization:** Pre-trained weights act as a strong prior, lowering the risk of overfitting on a smaller dataset.

Why Binary Classification?

While the dataset includes four stages (Non-Demented, Very Mild, Mild, and Moderate Demented), we focused on **binary classification** (Healthy vs. Diseased). This prioritizes **clinical relevance**, as the primary diagnostic goal is identifying the presence of Alzheimer's. To implement this, we merged all dementia stages into a single "Diseased" class during preprocessing.

4.1 Model Architectures Employed

To ensure a comprehensive evaluation, we employed three CNN architectures, comparing a custom baseline against two established transfer learning models with distinct computational profiles.

4.1.1 Custom CNN (Baseline)

Developed from scratch, this model serves as a reference point to measure the advantage of transfer learning over a network trained solely on the target dataset. Its compact architecture also allows us to test how effectively a lightweight model can learn disease patterns from limited medical data.

4.1.2 VGG16 (Transfer Learning)

VGG16 was chosen for its proven ability to extract rich hierarchical visual features. Its depth and stability make it ideal for capturing subtle structural abnormalities in brain MRIs. This serves as our primary high-capacity model for performance-oriented evaluation.

In addition, we developed an improved version by incorporating **SMOTE-based data balancing** and **hyperparameter tuning** via grid search.

4.1.3 MobileNetV2 (Lightweight CNN)

MobileNetV2 provides a lightweight alternative to evaluate the trade-off between predictive performance and computational efficiency. By using depthwise separable convolutions, it substantially reduces trainable parameters and costs. Including this model allows for a direct comparison between heavy, high-capacity architectures and resource-efficient ones.

4.2 Feature Engineering and Dimensionality Reduction

A key advantage of our CNN-based approach is that it **eliminates the need for manual feature engineering**.

Automatic Feature Learning: Traditional machine learning (e.g., SVM, Random Forest) requires expert-designed features like hippocampal volume or cortical thickness. In contrast, our CNNs learn the most relevant features directly from image pixels during training. This enables the discovery of novel or subtle biomarkers that manual extraction might overlook.

Dimensionality Reduction: We utilized a GlobalAveragePooling2D layer in all models as an implicit dimensionality reduction technique. By averaging each feature map, it converts high-dimensional 3D maps into a single 1D feature vector. This significantly reduces parameters before the final dense layers, preventing overfitting and accelerating computation without requiring explicit techniques like PCA.

5. Experiment

This section outlines the training and evaluation procedures, including data preprocessing, model configuration, hyperparameter tuning, and performance assessment.

5.1 Training Process

5.1.1 Normalization and Resizing

All MRI images were preprocessed prior to training. Pixel intensities were normalized from the range [0, 255] to [0, 1] by scaling with a factor of 1/255, which facilitates stable gradient-based optimization. Images were resized to **224 × 224 pixels** to conform to the input requirements of VGG16 and MobileNetV2.

5.1.2 Dataset Splitting (Training and Validation)

To ensure a robust estimate of model generalization, we used **patient-wise stratified 5-fold cross-validation**. Images were grouped by patient identifiers extracted from filenames. The 68 unique patients were partitioned into five stratified folds to preserve class distribution. In each of the five iterations, approximately 80% of patients were allocated for training and 20% for validation, with each fold serving as the validation set exactly once. This strategy prevents data leakage by ensuring all images from a single patient remain within a single fold.

Performance

Estimation:

Model performance is reported as the **mean across all validation folds** to estimate generalization to unseen patients. Due to the limited dataset size, a separate held-out test set was not utilized.

5.1.3 Data Augmentation

To enhance generalization and mitigate overfitting, data augmentation was applied exclusively to the training data through on-the-fly transformations.

For VGG16 and MobileNetV2:

horizontal and vertical flipping, rotation ($\pm 10\%$), zoom ($\pm 10\%$), contrast adjustment, and translation.

For the custom CNN:

moderate zoom, contrast variation, horizontal flipping, and small-angle rotation.

5.1.4 Regularization Techniques

Several regularization methods were incorporated to reduce overfitting:

Dropout: rates of 0.3

L2 Weight Regularization:

0.0005 (custom CNN) and 0.001 (VGG16 classifier head)

Batch Normalization: improves training stability and convergence

Early Stopping: training terminated upon stagnation of validation loss (patience = 4; 6 for MobileNetV2), with restoration of optimal weights

Class Imbalance Handling: class weights computed via `compute_class_weight` were applied during training

5.1.5 Training Configuration

All models were trained using the following configuration:

Optimizer: Adam

Loss Function: Binary Cross-Entropy (VGG16, MobileNetV2) and Binary Cross-Entropy (VGG16, MobileNetV2)

Batch Size: 32

Training Duration: up to convergence with early stopping

5.2 Model Architectures

5.2.1 Custom CNN (Baseline)

The Custom CNN was developed from scratch and consists of three convolutional blocks designed to progressively extract hierarchical image features.

Each convolutional block includes:

1. Conv2D layer with filter sizes of 32, 64, and 128, kernel size (3×3), and ‘same’ padding
2. Batch Normalization to stabilize training
3. ReLU activation function
4. MaxPooling2D for spatial downsampling

After feature extraction, the architecture includes:

1. GlobalAveragePooling2D to reduce high-dimensional feature maps into a compact feature vector
2. Dense layer (128 units, ReLU activation) with L2 regularization (0.0005) to prevent overfitting
3. Dropout layer (rate = 0.4) for additional regularization
4. Output layer (1 unit, sigmoid activation) for binary classification

This architecture is relatively lightweight and serves as a baseline for comparison with transfer learning models.

5.2.2 VGG16 (Transfer Learning)

The VGG16 model was initialized with **pre-trained ImageNet weights** using **include_top=False** to remove the original classification head.

Fine-tuning strategy:

- The majority of convolutional layers were **frozen**
- The **last 4 layers were unfrozen** to allow adaptation to MRI-specific features

The custom classification head added on top of VGG16 includes:

- **GlobalAveragePooling2D** to reduce feature map dimensionality
- **Batch Normalization** to improve convergence stability
- **Dense layer (128 units, ReLU activation)** with **L2 regularization (0.001)**
- **Dropout layer (rate = 0.5)** to reduce overfitting
- **Output layer (1 unit, sigmoid activation)**

Additionally, **data augmentation layers** (random flip, rotation, zoom, contrast, and translation) were integrated at the input stage of the model and applied only during training.

5.2.3 MobileNetV2 (Lightweight CNN)

MobileNetV2 was also initialized with **ImageNet pre-trained weights** and configured for transfer learning using **include_top=False**.

Fine-tuning strategy:

All layers were frozen except the **last 2 layers**, allowing limited adaptation to the target dataset while preserving learned representations

The classification head consists of:

1. **GlobalAveragePooling2D** for dimensionality reduction
2. **Dropout layer (rate = 0.5)** for regularization
3. **Output layer (1 unit, sigmoid activation)**

Similar to VGG16, **data augmentation layers** were applied at the input stage during training.

This architecture is significantly more efficient in terms of parameters and computation, making it suitable for evaluating the trade-off between model complexity and performance.

5.3 Hyperparameter Tuning

Hyperparameter optimization was conducted for the VGG16-based model using a **grid search** approach.

<i>Hyperparameter</i>	<i>Values Explored</i>	<i>Optimal Value</i>
<i>Learning Rate</i>	1e-4, 1e-5	1e-4
<i>Dropout Rate</i>	0.3 , 0.5	0.3

Table 5:Hyperparameter Tuning

Each configuration was evaluated using 5-fold cross-validation, and the optimal parameters were selected based on the highest mean validation accuracy.

5.4 Evaluation Metrics

Model performance was assessed using standard classification metrics derived from the confusion matrix:

- **Accuracy:** overall classification performance
- **Precision:** proportion of correct positive predictions

- **Recall (Sensitivity):** ability to identify diseased cases
- **F1-score:** harmonic mean of precision and recall
- **Confusion Matrix:** detailed breakdown of predictions

5.5 Libraries and Computational Resources

All experiments were conducted using **Google Colab** with the following resources:

- **GPU:** NVIDIA Tesla T4
- **CPU:** Intel Xeon

Software libraries:

- TensorFlow (Keras) for model development
- scikit-learn for data splitting and evaluation
- NumPy for numerical computations
- Matplotlib and Seaborn for visualization

6. Results and Discussion

This section presents and analyzes the performance of all models and explains how they were evaluated using 5-fold patient-wise cross-validation to ensure robust evaluation on unseen patients.

6.1 Overall Model Performance

Table 6 summarizes the overall performance of all models across the five cross-validation folds.

MODEL	ACCURACY	PRECISION (H / D)	RECALL (H / D)	F1-SCORE (H / D)
CUSTOM CNN	0.80	0.69 / 0.85	0.60 / 0.89	0.64 / 0.87
MOBILENETV2	0.60	0.37 / 0.78	0.55 / 0.63	0.44 / 0.69
STANDARD VGG16	0.75	0.58 / 0.83	0.56 / 0.83	0.57 / 0.83
ENHANCED VGG16 (SMOTE + GRID SEARCH)	0.78	0.62 / 0.86	0.66 / 0.83	0.64 / 0.85

Table 6: Overall 5-Fold Cross-Validation Performance

Note: H = Healthy, D = Diseased

Due to class imbalance, balanced evaluation metrics such as recall and F1-score were prioritized over accuracy alone.

6.2 Model Comparison

Custom CNN: Achieved the highest overall accuracy (80%) and the strongest diseased recall (0.89). However, the model showed a bias toward the diseased class, as reflected by the relatively low healthy recall (0.60), indicating increased false positives.

Enhanced VGG16 (SMOTE & Grid Search): Delivered the most balanced overall performance across both classes. Although its accuracy (78%) was slightly lower than the Custom CNN, it achieved better class balance, stronger healthy recall, and more stable generalization, making it the most clinically suitable model.

Standard VGG16: Reached 75% accuracy with a diseased recall of 0.83. However, it demonstrated a clear bias, with healthy recall dropping to 0.56, making it less effective than the tuned version.

MobileNetV2: The weakest architecture, achieving only 60% accuracy. With a diseased recall of 0.63 and low healthy precision (0.37), this lightweight model lacked the complexity required to capture fine-grained MRI biomarkers.

6.3 Generalization to Unseen Patients

To ensure clinical reliability, we employed patient-wise cross-validation, which confined all images from a single patient to one fold. This prevented data leakage and provided a more realistic estimate of how the models perform on previously unseen individuals.

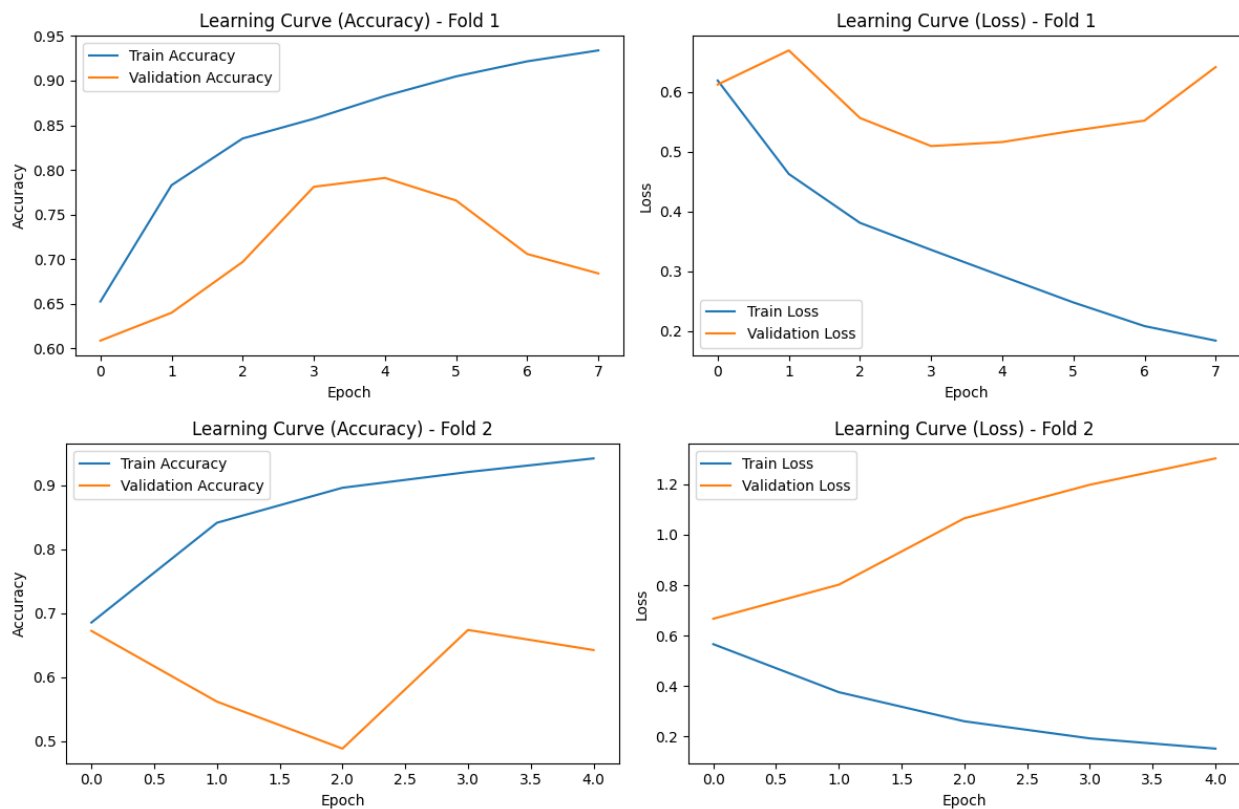
The VGG16 architectures (Standard and Enhanced) showed the most stable generalization. The combination of transfer learning and regularization techniques effectively mitigated overfitting, maintaining reliable feature extraction across different patient groups. In contrast, MobileNetV2 exhibited signs of underfitting; its validation performance plateaued early even as training performance improved, confirming that its simplified architecture could not generalize the structural patterns present in MRI scans.

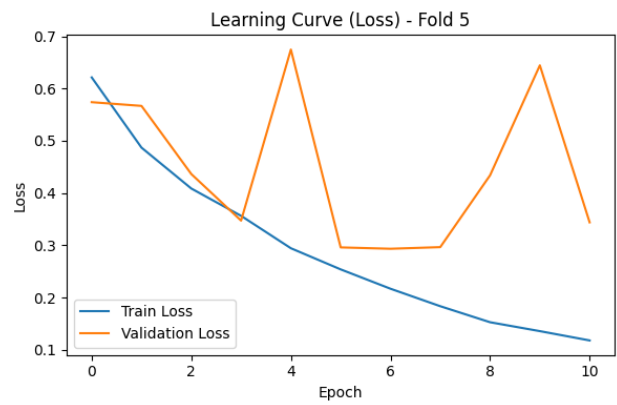
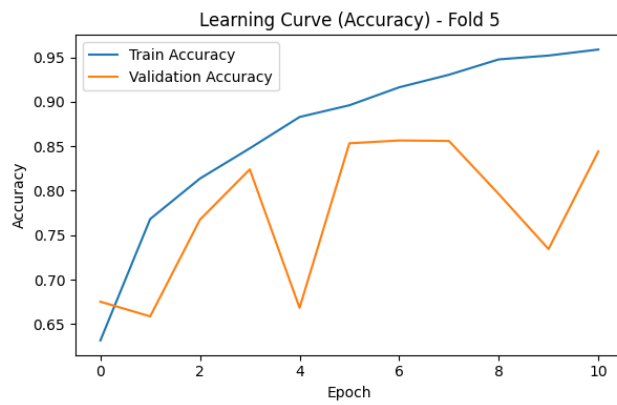
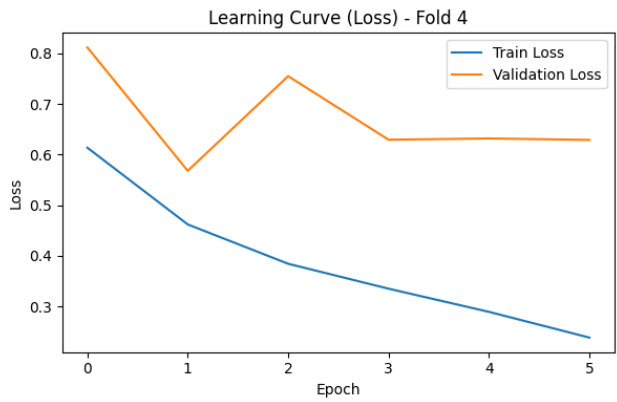
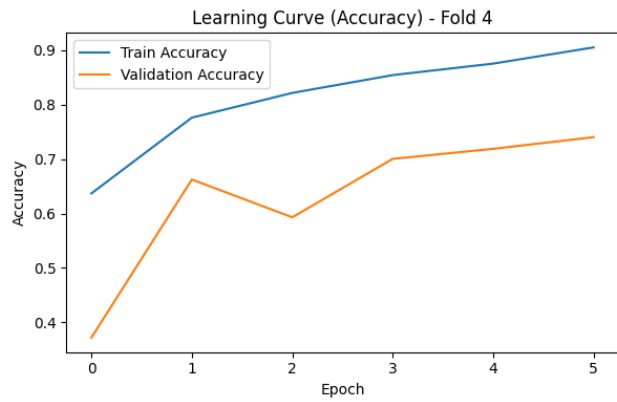
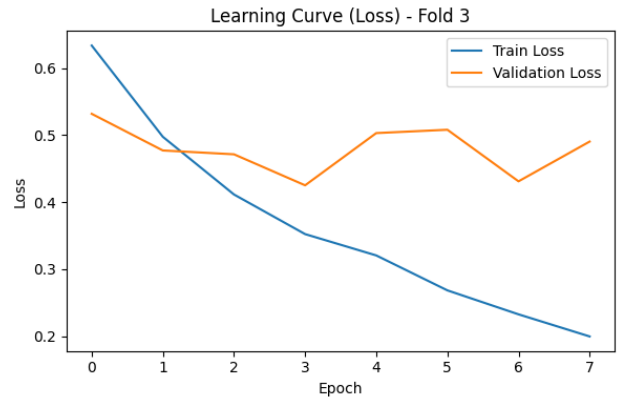
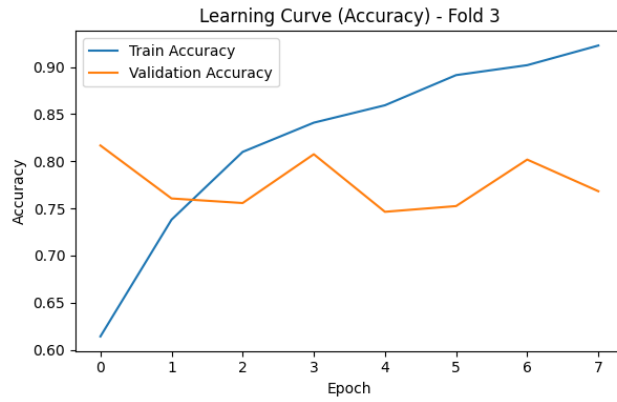
Overall, transfer learning models demonstrated stronger balance and generalization, while the Custom CNN achieved the highest raw classification accuracy.

6.4 Models Analysis

Figures 2 through 5 present the confusion matrices and learning curves for each model.

Standard VGG16 Confusion Matrix and Learning Curves:





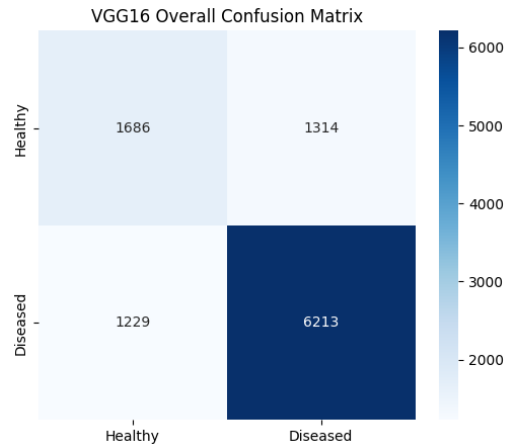


Figure 2: Standard VGG16 Confusion Matrix

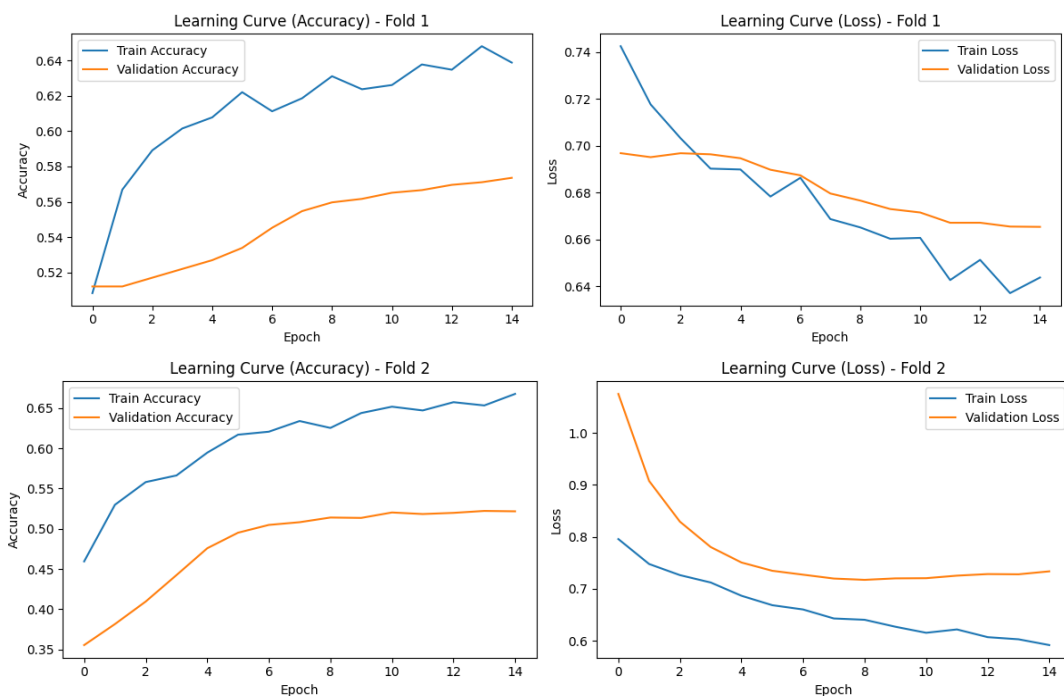
Overall Classification Report:

	<i>precision</i>	<i>recall</i>	<i>f1-score</i>	<i>support</i>
<i>Healthy</i>	0.58	0.56	0.57	3000
<i>Diseased</i>	0.83	0.83	0.83	7442

Overall Accuracy: 0.75

Macro Average F1-score : 0.70

MobileNetV2 Confusion Matrix and Learning Curves:



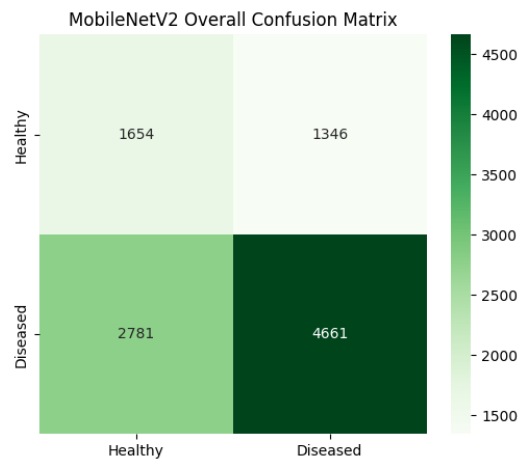
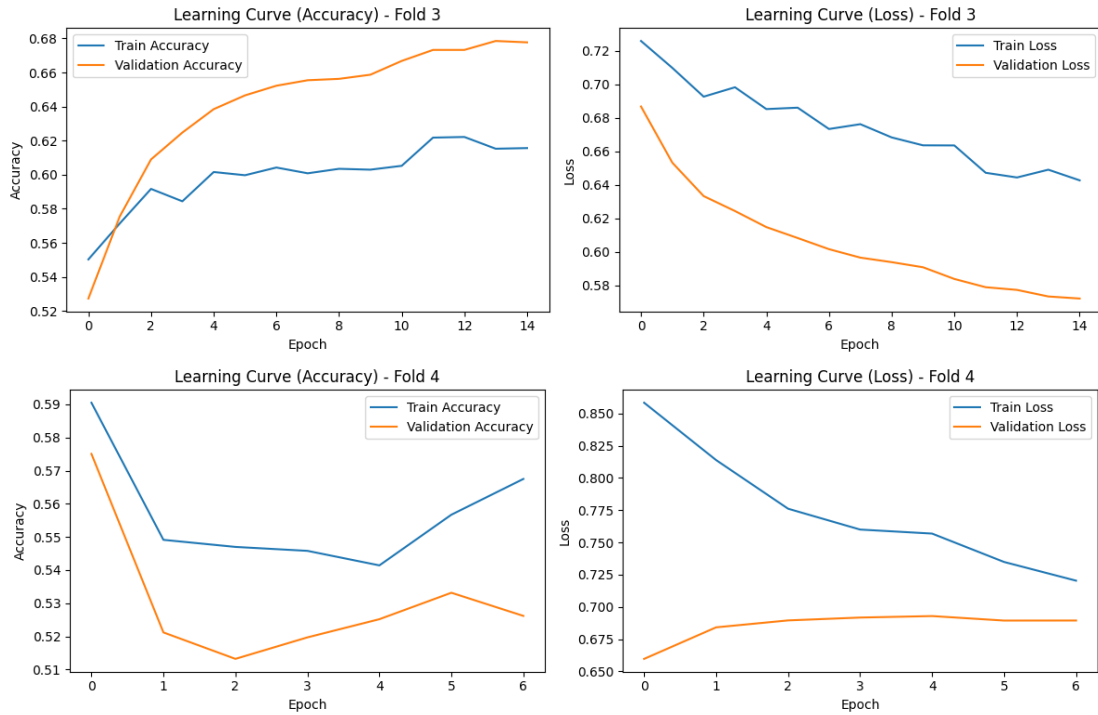


Figure 3: MobileNetV2 Confusion Matrix

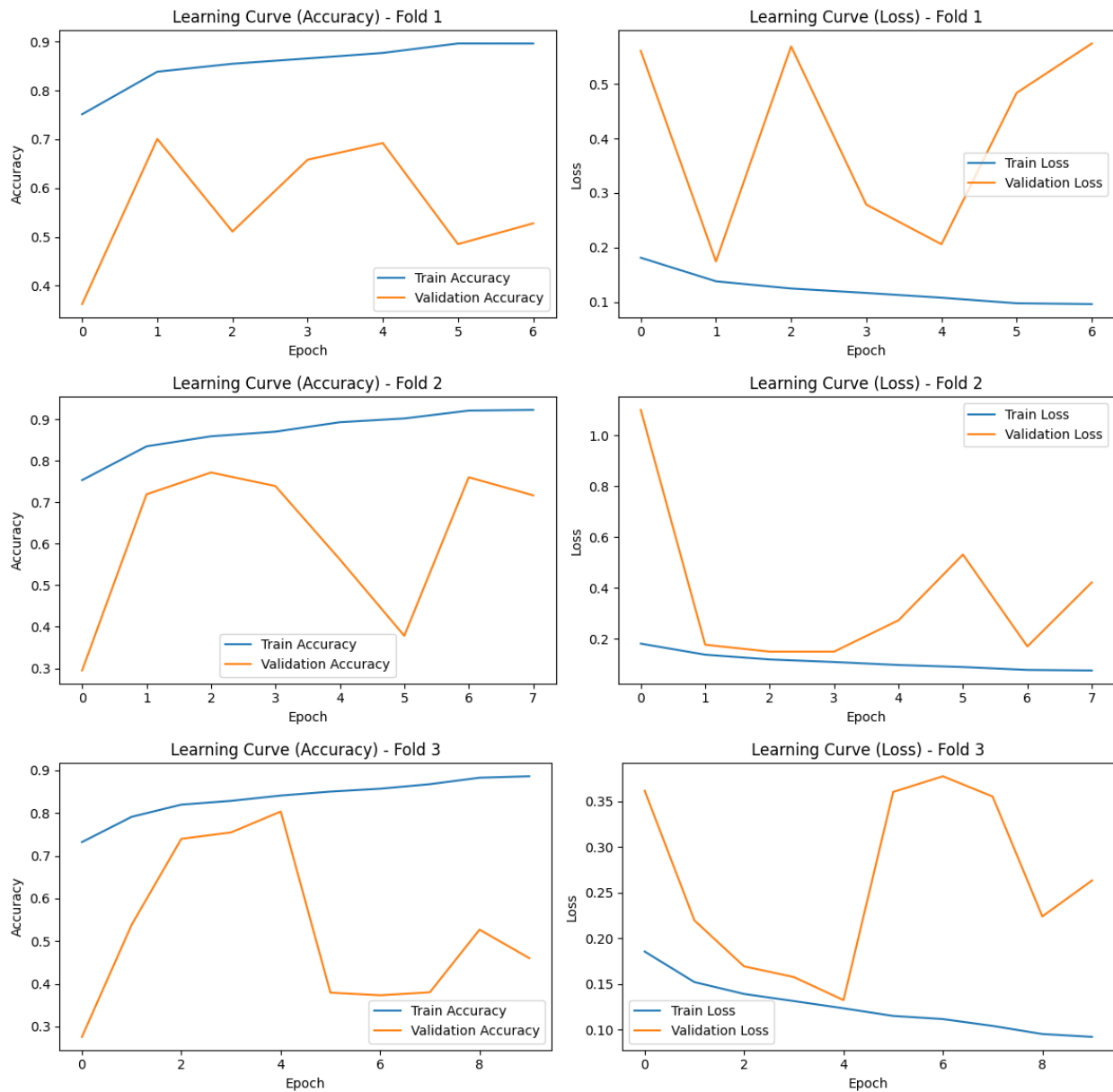
Overall Classification Report:

	precision	recall	f1-score	support
Healthy	0.37	0.55	0.44	3000
Diseased	0.78	0.63	0.69	7442

Overall Accuracy: 0.60

Macro Average F1-score : 0.57

Custom CNN Confusion Matrix and Learning Curves:



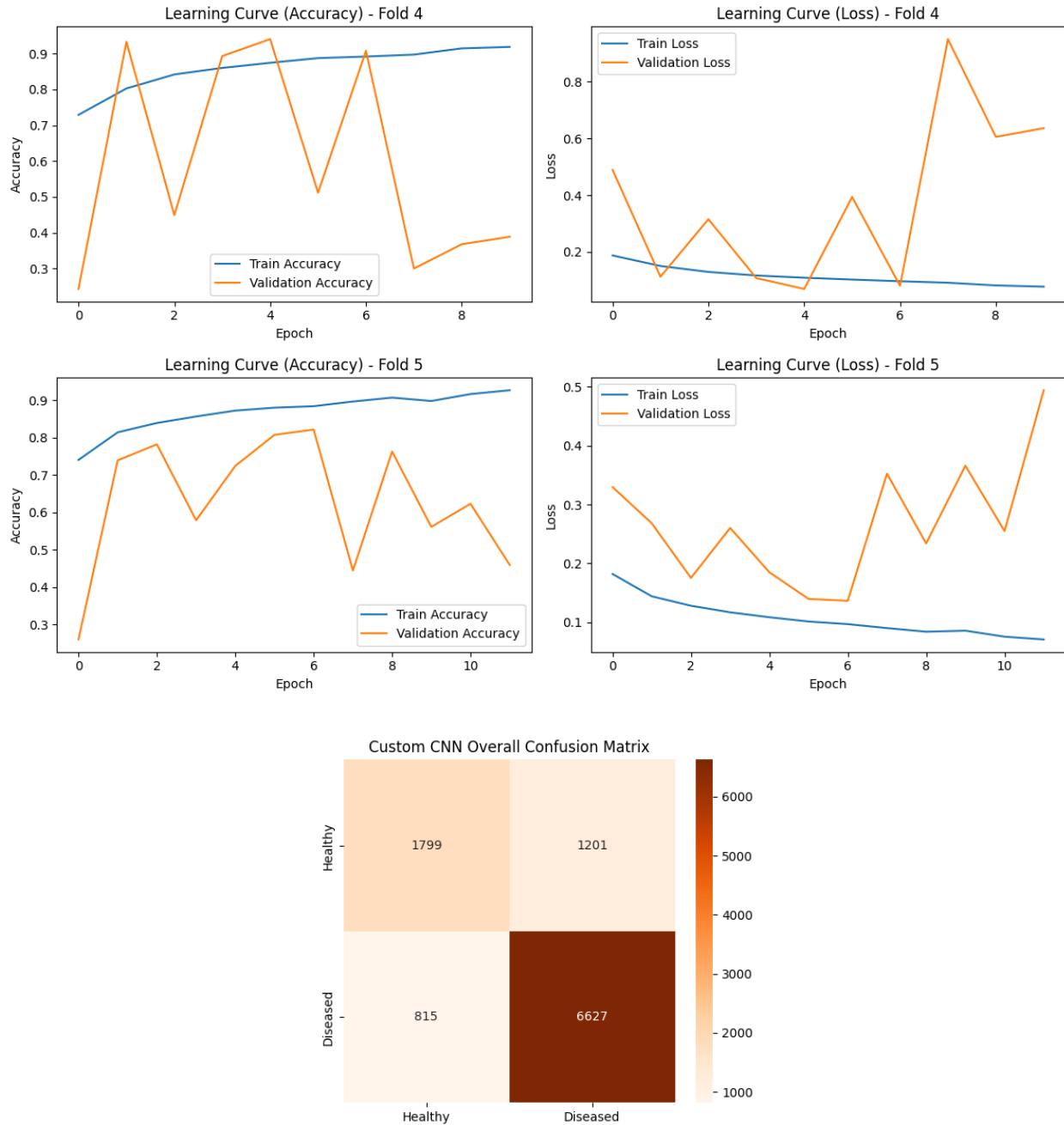


Figure 4: Custom CNN Confusion Matrix

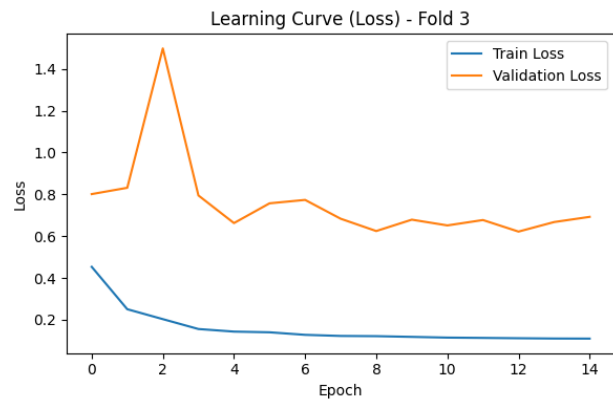
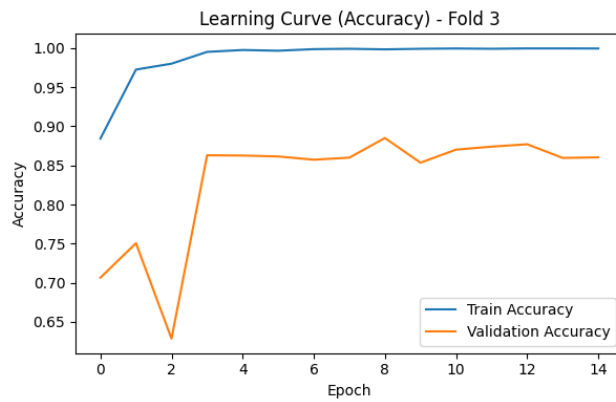
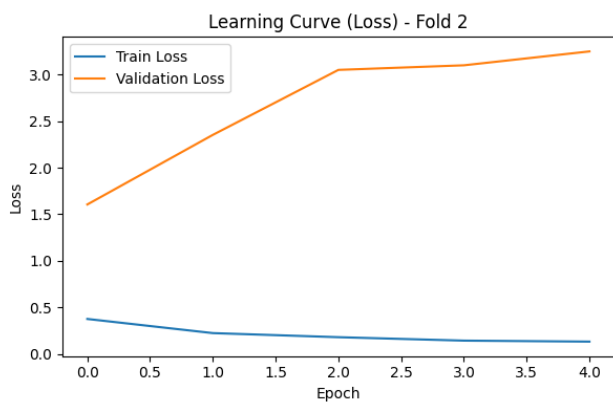
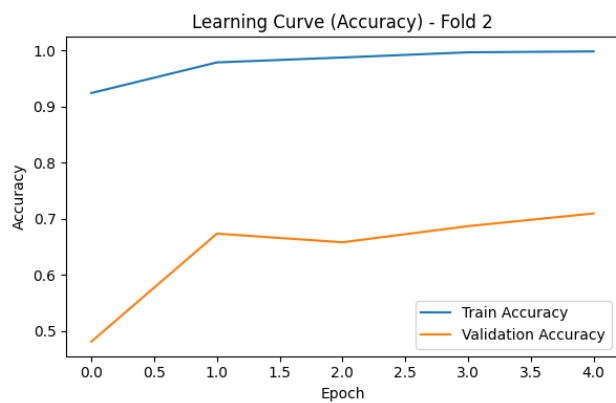
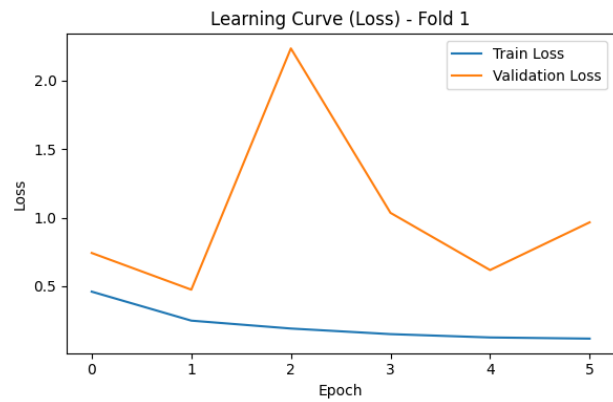
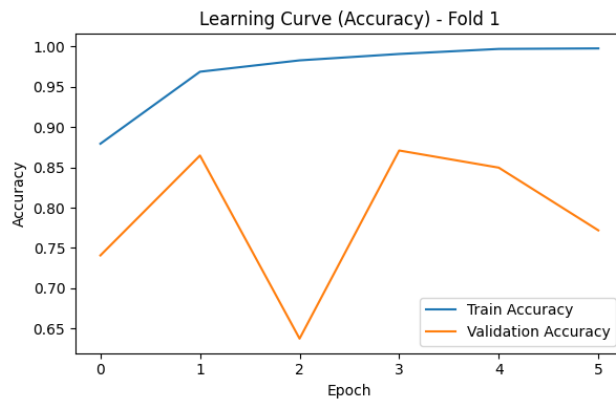
Overall Classification Report:

	<i>precision</i>	<i>recall</i>	<i>f1-score</i>	<i>support</i>
<i>Healthy</i>	0.69	0.60	0.64	3000
<i>Diseased</i>	0.85	0.89	0.87	7442

Overall Accuracy: 0.80

Macro Average F1-score : 0.75

Enhanced VGG16 Confusion Matrix and Learning Curves:



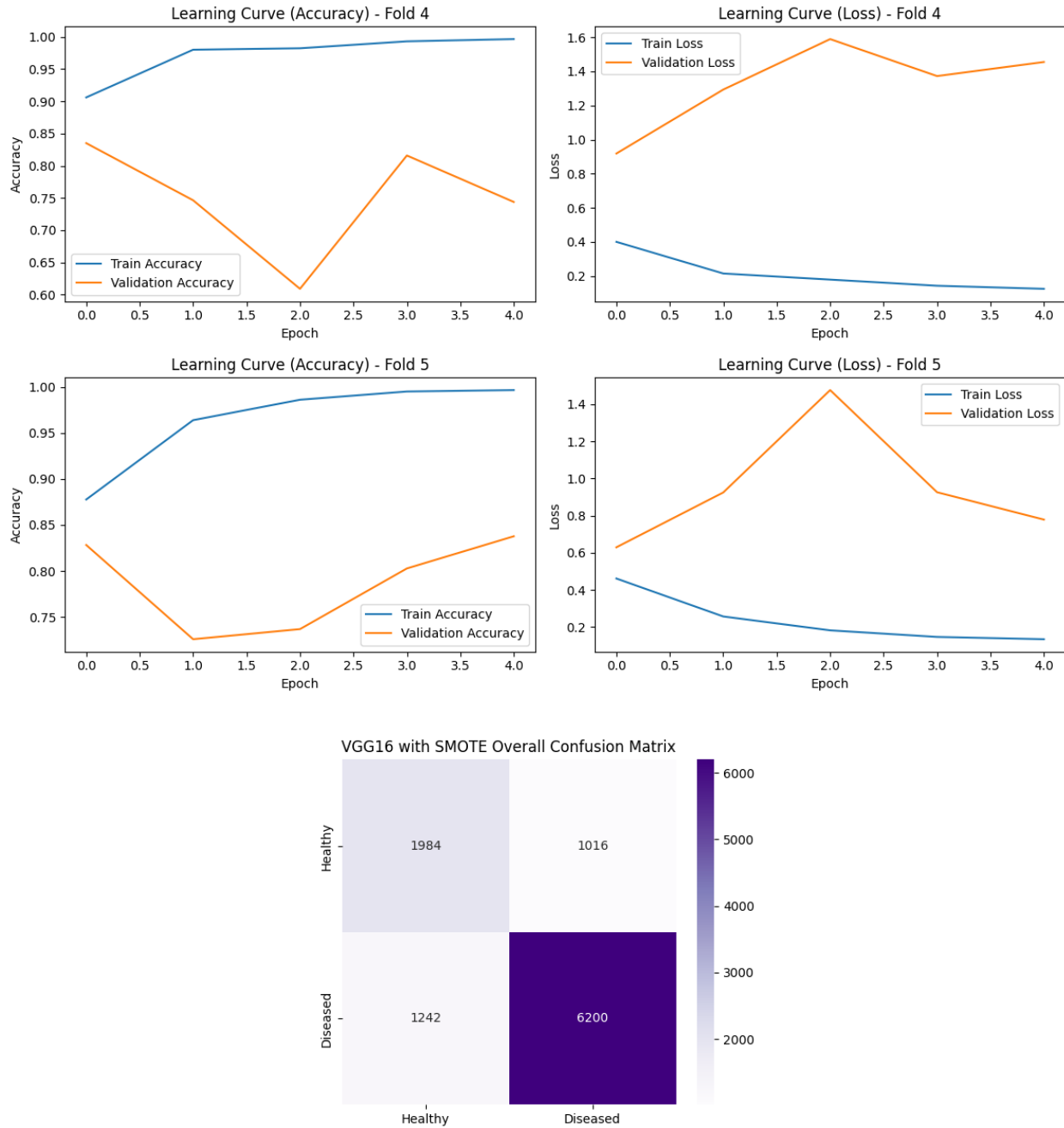


Figure 5: Enhanced VGG16 Confusion Matrix

Overall Classification Report:

	precision	recall	f1-score	support
Healthy	0.62	0.66	0.64	3000
Diseased	0.86	0.83	0.85	7442

Overall Accuracy: 0.78

Macro Average F1-score : 0.74

Best Config → LR: 0.0001, Dropout: 0.3

A consistent trend across all models is the higher misclassification rate of healthy subjects compared to diseased subjects, suggesting that distinguishing normal anatomical variation from early pathological changes remains challenging in MRI-based classification.

6.5 Interpretation of Results

The experimental outcomes provide several critical insights into the application of deep learning for neuroimaging:

Synergy of Transfer Learning and Optimization: While transfer learning provides a robust starting point, The improvement from the Standard VGG16 (75%) to the Enhanced VGG16 (78%) demonstrates the effectiveness of SMOTE-based balancing and hyperparameter tuning.

The Architecture Threshold: The limited performance of MobileNetV2 indicates a "complexity floor" for Alzheimer's detection. Unlike standard object recognition where lightweight models excel, the subtle, spatially distributed biomarkers of dementia (such as localized cortical thinning) require the higher representational depth found in larger architectures like VGG16.

The Biological "Decision Boundary" Challenge: A persistent difficulty across all models was healthy class precision. This reveals a fundamental challenge in medical imaging: the anatomical "gray area." Normal age-related brain changes can structurally mimic early pathological atrophy, creating an overlapping feature distribution that makes confirming the absolute absence of disease more difficult than detecting its presence.

Validation Integrity: The consistency of results across the patient-wise 5-fold cross-validation suggests that the models learned generalizable pathological features rather than memorizing patient-specific noise. This validates the methodology's potential for clinical decision-support tools where generalization to new, unseen patients is the primary requirement.

7. Conclusion

This project demonstrated the application of deep learning for classifying Alzheimer's disease from brain MRI scans. Multiple CNN architectures were systematically evaluated, including a custom CNN and transfer learning models such as VGG16 and MobileNetV2.

The results show that the **Custom CNN achieved the highest overall accuracy (80%)**, but its performance was slightly biased toward the diseased class, as reflected by lower healthy recall (0.60) compared to diseased recall (0.89). This indicates strong sensitivity for detecting Alzheimer's cases, but reduced balance across classes.

The **Enhanced VGG16 with SMOTE and Grid Search** demonstrated the most **balanced and clinically suitable performance**, achieving **78% accuracy** with strong diseased-class metrics (precision: 0.86, recall: 0.83, F1-score: 0.85). The use of SMOTE, dropout tuning, batch normalization, and L2 regularization improved robustness and reduced class imbalance effects, making it more suitable for real-world clinical scenarios.

The **Standard VGG16** achieved moderate performance (75% accuracy), but showed lower sensitivity for the healthy class (recall: 0.56), indicating a bias toward diseased predictions and less stability compared to the enhanced version.

In contrast, **MobileNetV2 performed the worst (60% accuracy)**, with low precision and F1-score for the healthy class. This confirms that its lightweight architecture was insufficient for capturing the complex patterns in MRI-based Alzheimer's classification, leading to underfitting and poor generalization.

Overall, the study highlights that while deeper or pretrained models provide strong baselines, **performance is significantly improved through class balancing and regularization techniques**. The Enhanced VGG16 emerges as the most clinically suitable model due to its balanced trade-off between accuracy, sensitivity, and generalization.

One limitation of this study is the relatively small number of unique patients (68), which may limit broader generalization despite the use of patient-wise cross-validation. Future work could explore ensemble learning, more advanced imbalance-handling strategies, and further hyperparameter optimization to improve robustness and potentially exceed current performance levels.

8. Contributions

The project was developed collaboratively by all team members. Each member's specific contributions are detailed below:

Noura Alamro (444200941):

- Implemented and trained the **Custom CNN** , including architecture design, configuring data augmentation layers, setting training parameters, regularization and evaluating performance.
- Implemented and trained the **VGG16 models** (both standard and enhanced versions), including SMOTE integration and grid search over hyperparameters.
- Reviewed the related work by Thevar et al. [5]

Shooq Alawadah (444201083):

- Researched, selected, and documented the **OASIS dataset**, including analyzing its characteristics, class distribution, and suitability for the task.
- Implemented and trained the **Custom CNN** model, including architecture design and regularization (BatchNorm, L2, Dropout, Focal Loss).
- Wrote the **Experiment** and **Methods** sections, documenting the training process, hyperparameter tuning, and evaluation metrics.
- Reviewed the related work by Hossain et al. [6], incorporating findings into the report.
- Wrote and formatted the **complete final report**, ensuring all sections met the required structure and formatting guidelines.
- Formatted the **complete final presentation**.

Reema Alraqibah (444201069):

- Implemented and trained the **Custom CNN** model, including architecture design and regularization (BatchNorm, L2, Dropout, Focal Loss).
- Verified the dataset was well-organized, balanced, and met the project's technical standards before experiments.
- Reviewed the related work Mohammed et al. [7] on hybrid deep learning with SVM

- Conducted comparative analysis of model results and contributed to the **Results and Discussion** section.
- Formatted the **complete final presentation**.

Ragad Alrashid (444200498):

- Implemented and trained the **VGG16** model and **MobileNetV2**.
- Reviewed the related work Hu et al.[8]. for Alzheimer's stage classification using hippocampal MRI segmentation.
- Extracted the models for the **live demo dataset**.

9. References

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